

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Set Notation & Venn Diagrams

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

SET NOTATION & VENN DIAGRAMS · CH1 · L2



ENG. ESLAM AHMED

The Map

This lesson collapses Set Notation & Venn Diagrams into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Fill the intersection first

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Use union and intersection

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Use complements

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Read probability from the diagram

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Calculate statistic

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 23 website questions, 23 checked solutions, 3 local PDFs read.

01 Fill the intersection first

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Set Notation & Venn Diagrams question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$B = \{b, l, u, e\}, \quad G = \{g, r, e, y\}, \quad W = \{w, h, i, t, e\}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q4

Calculate value

$$B = \{b, l, u, e\}, \quad G = \{g, r, e, y\}, \quad W = \{w, h, i, t, e\}$$

means all the letters that

$$B \cup G = \{b, l, u, e, g, r, y\}$$

Finish: (a)(i) $\{b, l, u, e, g, r, y\}$, (a)(ii) $\{w, h, i, t\}$. Serena is incorrect.

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

4. $B = \{b, l, u, e\}$ $G = \{g, r, e, y\}$ $W = \{w, h, i, t, e\}$ (a) List all the members of the set (i) $B \cup G$ (ii) $W \cap G'$ (2) Serena writes down the statement $B \cap G \cap W = \text{empty set}$ (b) Is Serena's statement correct? You must give a reason for your answer. (1)

YOUR SOLUTION

02 Use union and intersection

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Set Notation & Venn Diagrams question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$G \cap S \text{ only} = 11 - 6 = 5$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Specimen 4WM1H Q14

Place the centre value

This is a reasoning step: write one clear sentence, then continue the calculation.

Fill the pair-only regions

$$G \cap S \text{ only} = 11 - 6 = 5$$

Finish: Venn regions: G only 13, S only 9, C only 24, $G \cap S$ only 5, $S \cap C$ only 11, $G \cap C$ only 12, all three 6, none 20. Also 36, 44, 35, and $\frac{18}{53}$.

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 02.

QUESTION

14 100 farmers are asked if they have goats, sheep or chickens. Complete the Venn diagram, find $n(G)$, $n((G \cup S)')$, $n(G' \cap C)$, and the probability a farmer with chickens also has goats.

YOUR SOLUTION

03 Use complements

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Set Notation & Venn Diagrams question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\mathcal{E} = \{20, 21, 22, 23, 24, 25, 26, 27, 28, 29\}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1H Q4

Calculate value

$$\mathcal{E} = \{20, 21, 22, 23, 24, 25, 26, 27, 28, 29\}$$

The odd numbers are

$$A = \{21, 23, 25, 27, 29\}$$

Finish: (i) $\{21, 27\}$, (ii) $\{21, 23, 24, 25, 27, 29\}$

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 03.

QUESTION

4 E = {20, 21, 22, 23, 24, 25, 26, 27, 28, 29} A = {odd numbers} B = {multiples of 3} List the members of the set (i) A intersection B (1) (ii) A union B (1)

YOUR SOLUTION

04 Read probability from the diagram

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Set Notation & Venn Diagrams question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$G \cap S \text{ only} = 11 - 6 = 5$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Specimen 4WM1H Q14

Place the centre value

This is a reasoning step: write one clear sentence, then continue the calculation.

Fill the pair-only regions

$$G \cap S \text{ only} = 11 - 6 = 5$$

Finish: Venn regions: G only 13, S only 9, C only 24, $G \cap S$ only 5, $S \cap C$ only 11, $G \cap C$ only 12, all three 6, none 20. Also 36, 44, 35, and $\frac{18}{53}$.

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 04.

QUESTION

14 100 farmers are asked if they have goats, sheep or chickens. Complete the Venn diagram, find $n(G)$, $n((G \cup S)')$, $n(G' \cap C)$, and the probability a farmer with chickens also has goats.

YOUR SOLUTION

05 Calculate statistic

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Set Notation & Venn Diagrams question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$B = \{b, l, u, e\}, \quad G = \{g, r, e, y\}, \quad W = \{w, h, i, t, e\}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q4

Calculate value

$$B = \{b, l, u, e\}, \quad G = \{g, r, e, y\}, \quad W = \{w, h, i, t, e\}$$

means all the letters that

$$B \cup G = \{b, l, u, e, g, r, y\}$$

Finish: (a)(i) $\{b, l, u, e, g, r, y\}$, (a)(ii) $\{w, h, i, t\}$. Serena is incorrect.

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

4. $B = \{b, l, u, e\}$ $G = \{g, r, e, y\}$ $W = \{w, h, i, t, e\}$ (a) List all the members of the set (i) $B \cup G$ (ii) $W \cap G'$ (2) Serena writes down the statement $B \cap G \cap W = \text{empty set}$ (b) Is Serena's statement correct? You must give a reason for your answer. (1)

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Fill the intersection first
"work out", "show", "hence", a familiar exam phrase	Use union and intersection
"work out", "show", "hence", a familiar exam phrase	Use complements
"work out", "show", "hence", a familiar exam phrase	Read probability from the diagram
"work out", "show", "hence", a familiar exam phrase	Calculate statistic

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the

trigger, write the first line, and let the working follow.